# Results — Etihad BW-BI vs TAT WaterLine

Annual fuel savings, side by side · Annualized ×4 from Jun–Aug 2026 (S26) · 526 routes from Jorn Fugger (EY Ops Engineering)

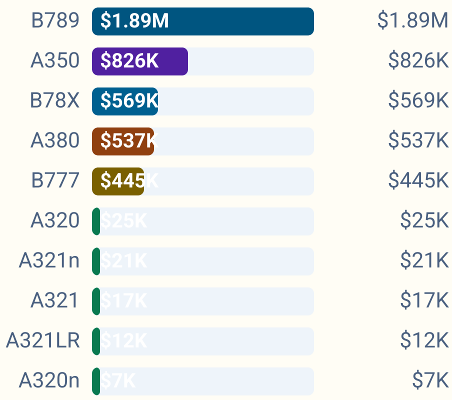
## EY Etihad BW-BI Policy

**$4.35M**

5422.2 ton JET-A/yr · annualized ×4

| ELECTRONICALLY MONITORED | RESIDUE COST | JET-A SAVED |
|---|---|---|
| **$3.28M** | **$701K** | **6.78M L/yr** |
| A350 + B787 · 164 routes · auto-recorded by EY | water carried unused | 5422.2 ton/yr |

**SAVINGS BY FLEET**




## TAT TAT WaterLine *Smart Water. Real Results.*

**$9.78M**

$9.12M fuel + $664K CORSIA · annualized ×4

| TAT FUEL SAVINGS | CORSIA | $CO_2$ AVOIDED |
|---|---|---|
| **$9.12M** | **$664K** | **28,887** |
| WaterLine only | est. rate | ton/yr |

**SAVINGS BY FLEET**

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# Top 20 Routes — Annual Savings

Ranked by savings USD/yr · Etihad BW-BI (left) and TAT WaterLine (right)

## EY Top 20 — Etihad BW-BI Savings/yr

| Route | Aircraft | Savings/yr |
|---|---|---|
| JFK → AUH | A350 | $151K |
| AUH → JFK | A350 | $130K |
| LHR → AUH | A380 | $128K |
| AUH → LHR | A380 | $108K |
| BKK → AUH | A350 | $107K |
| CDG → AUH | A380 | $102K |
| AUH → BKK | A350 | $86K |
| AUH → CDG | A380 | $85K |
| ORD → AUH | B789 | $79K |
| BKK → AUH | B777 | $67K |
| YYZ → AUH | A380 | $66K |
| HKT → AUH | B78X | $63K |
| FRA → AUH | B789 | $59K |
| AUH → SYD | A350 | $56K |
| MNL → AUH | B777 | $55K |
| MLE → AUH | B789 | $54K |
| KUL → AUH | B789 | $53K |
| AUH → FRA | B789 | $49K |
| MUC → AUH | B789 | $48K |
| AMS → AUH | A350 | $46K |

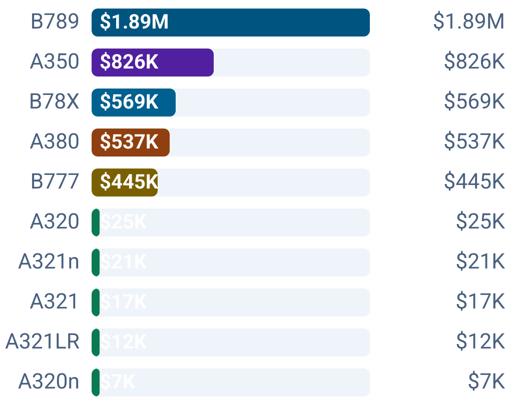
## TAT Top 20 — TAT WaterLine Savings/yr

| Route | Aircraft | Savings/yr |
|---|---|---|
| LHR → AUH | A380 | $279K |
| JFK → AUH | A350 | $256K |
| AUH → LHR | A380 | $224K |
| AUH → JFK | A350 | $209K |
| CDG → AUH | A380 | $168K |
| SYD → AUH | A350 | $158K |
| AUH → CDG | A380 | $135K |
| BKK → AUH | A350 | $128K |
| ATL → AUH | A350 | $123K |
| AUH → SYD | A350 | $121K |
| ICN → AUH | A350 | $116K |
| YYZ → AUH | A380 | $109K |
| BKK → AUH | B777 | $107K |
| MNL → AUH | B777 | $104K |
| AUH → BKK | A350 | $102K |
| ORD → AUH | B789 | $102K |
| AUH → ATL | A350 | $100K |
| NRT → AUH | A380 | $98K |
| KUL → AUH | B789 | $88K |
| AUH → BKK | B777 | $86K |

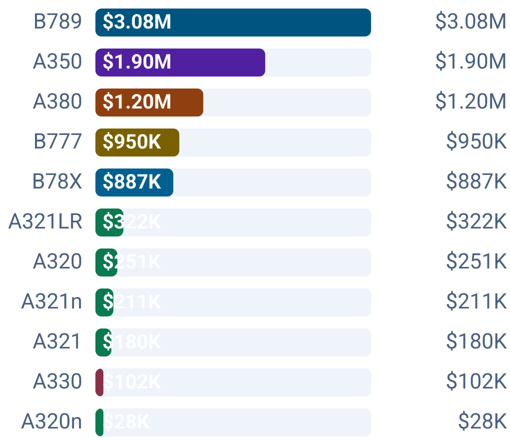
## Savings by Aircraft Family

Annual savings grouped by fleet family · Etihad BW-BI (left) and TAT WaterLine (right)

**EY** Savings by Family — Etihad BW-BI




**TAT** Savings by Family — TAT WaterLine




| FAMILY | ROUTES | FLIGHTS/YR | EY SAVINGS/YR | TAT SAVINGS/YR |
|---|---|---|---|---|
| B789 | 106 | 25,500 | $1.89M | $3.31M |
| B78X | 52 | 7,888 | $569K | $951K |
| A350 | 28 | 7,140 | $826K | $2.04M |
| A380 | 10 | 4,412 | $537K | $1.29M |
| B777 | 42 | 5,824 | $445K | $1.02M |
| A321LR | 70 | 17,896 | $12K | $346K |
| A321n | 80 | 13,020 | $21K | $226K |
| A321 | 46 | 13,340 | $17K | $193K |
| A320n | 18 | 2,220 | $7K | $30K |
| A320 | 70 | 18,456 | $25K | $269K |
| A330 | 4 | 1,068 | — | $110K |

# Results by Fleet — Etihad BW-BI

Per-fleet breakdown · routes, flights, tank, COW, current BW-BI policy and annual savings

### B789

| | |
|---|---|
| Routes | 106 |
| Flights/yr | 25,500 |
| Tank | 1,020L |
| COW | 0.2072 |
| BW-BI | Rev221 |
| Savings/yr | $1.89M |
| Residue/yr | $414K |

43.4% of total EY savings

### B78X

| | |
|---|---|
| Routes | 52 |
| Flights/yr | 7,888 |
| Tank | 1,020L |
| COW | 0.1817 |
| BW-BI | Rev98 |
| Savings/yr | $569K |
| Residue/yr | $124K |

13.1% of total EY savings

### A350

| | |
|---|---|
| Routes | 28 |
| Flights/yr | 7,140 |
| Tank | 1,500L |
| COW | 0.3015 |
| BW-BI | Rev35 |
| Savings/yr | $826K |
| Residue/yr | $163K |

19.0% of total EY savings

### A380

| | |
|---|---|
| Routes | 10 |
| Flights/yr | 4,412 |
| Tank | 2,266L |
| COW | 0.2683 |
| BW-BI | — |
| Savings/yr | $537K |

12.3% of total EY savings

### B777

| | |
|---|---|
| Routes | 42 |
| Flights/yr | 5,824 |
| Tank | 1,304L |
| COW | 0.2092 |
| BW-BI | Rev238 |
| Savings/yr | $445K |

10.2% of total EY savings

### A321LR

| | |
|---|---|
| Routes | 70 |
| Flights/yr | 17,896 |
| Tank | 200L |
| COW | 0.1304 |
| BW-BI | Rev18 |
| Savings/yr | $12K |

0.3% of total EY savings

### A321n

| | |
|---|---|
| Routes | 80 |
| Flights/yr | 13,020 |
| Tank | 200L |
| COW | 0.1304 |
| BW-BI | Rev26 |
| Savings/yr | $21K |

0.5% of total EY savings

### A321

| | |
|---|---|
| Routes | 46 |
| Flights/yr | 13,340 |
| Tank | 200L |
| COW | 0.1304 |
| BW-BI | Rev140 |
| Savings/yr | $17K |

0.4% of total EY savings

### A320n

| | |
|---|---|
| Routes | 18 |
| Flights/yr | 2,220 |
| Tank | 200L |
| COW | 0.1098 |
| BW-BI | Rev10 |
| Savings/yr | $7K |

0.2% of total EY savings

### A320

| | |
|---|---|
| Routes | 70 |
| Flights/yr | 18,456 |
| Tank | 200L |
| COW | 0.1098 |
| BW-BI | Rev226 |
| Savings/yr | $25K |

0.6% of total EY savings

### A330

| | |
|---|---|
| Routes | 4 |
| Flights/yr | 1,068 |
| Tank | 700L |
| COW | 0.3015 |
| BW-BI | — |
| Savings/yr | — |

0.0% of total EY savings

# Results by Fleet — TAT WaterLine

Per-fleet breakdown · routes, flights, tank, COW, TAT savings and CORSIA

### B789

| | |
|---|---|
| Routes | **106** |
| Flights/yr | **25,500** |
| Tank | **1,020L** |
| COW | **0.2072** |
| TAT Savings/yr | **$3.08M** |
| CORSIA/yr | **$224K** |
| Total/yr | **$3.31M** |

33.8% of total TAT

### B78X

| | |
|---|---|
| Routes | **52** |
| Flights/yr | **7,888** |
| Tank | **1,020L** |
| COW | **0.1817** |
| TAT Savings/yr | **$887K** |
| CORSIA/yr | **$65K** |
| Total/yr | **$951K** |

9.7% of total TAT

### A350

| | |
|---|---|
| Routes | **28** |
| Flights/yr | **7,140** |
| Tank | **1,500L** |
| COW | **0.3015** |
| TAT Savings/yr | **$1.90M** |
| CORSIA/yr | **$138K** |
| Total/yr | **$2.04M** |

20.9% of total TAT

### A380

| | |
|---|---|
| Routes | **10** |
| Flights/yr | **4,412** |
| Tank | **2,266L** |
| COW | **0.2683** |
| TAT Savings/yr | **$1.20M** |
| CORSIA/yr | **$87K** |
| Total/yr | **$1.29M** |

13.2% of total TAT

### B777

| | |
|---|---|
| Routes | **42** |
| Flights/yr | **5,824** |
| Tank | **1,304L** |
| COW | **0.2092** |
| TAT Savings/yr | **$950K** |
| CORSIA/yr | **$69K** |
| Total/yr | **$1.02M** |

10.4% of total TAT

### A321LR

| | |
|---|---|
| Routes | **70** |
| Flights/yr | **17,896** |
| Tank | **200L** |
| COW | **0.1304** |
| TAT Savings/yr | **$322K** |
| CORSIA/yr | **$23K** |
| Total/yr | **$346K** |

3.5% of total TAT

### A321n

| | |
|---|---|
| Routes | **80** |
| Flights/yr | **13,020** |
| Tank | **200L** |
| COW | **0.1304** |
| TAT Savings/yr | **$211K** |
| CORSIA/yr | **$15K** |
| Total/yr | **$226K** |

2.3% of total TAT

### A321

| | |
|---|---|
| Routes | **46** |
| Flights/yr | **13,340** |
| Tank | **200L** |
| COW | **0.1304** |
| TAT Savings/yr | **$180K** |
| CORSIA/yr | **$13K** |
| Total/yr | **$193K** |

2.0% of total TAT

### A320n

| | |
|---|---|
| Routes | **18** |
| Flights/yr | **2,220** |
| Tank | **200L** |
| COW | **0.1098** |
| TAT Savings/yr | **$28K** |
| CORSIA/yr | **$2K** |
| Total/yr | **$30K** |

0.3% of total TAT

## A320

| | |
|---|---|
| Routes | 70 |
| Flights/yr | 18,456 |
| Tank | 200L |
| COW | 0.1098 |
| TAT Savings/yr | $251K |
| CORSIA/yr | $18K |
| Total/yr | $269K |

2.7% of total TAT

## A330

| | |
|---|---|
| Routes | 4 |
| Flights/yr | 1,068 |
| Tank | 700L |
| COW | 0.3015 |
| TAT Savings/yr | $102K |
| CORSIA/yr | $7K |
| Total/yr | $110K |

1.1% of total TAT

# WaterLine Validador

Lavatory physical-capacity cross-validation · narrowbody fleet · 400 mL/use · 4 min/use · 3 lavatories

Narrowbody fleet analysed A320-200 · A320neo · A321-200 · A321neo · A321LR Short & medium-haul stages · where lavatory capacity matters most

💡 How to read this matrix

Each cell answers one question: **on this flight type, can the lavatory physically consume all the water TAT loads?** Coverage % = water the lavatory can use ÷ water TAT boards. **Assumptions: 400 mL per use, 4 min per use, 3 lavatories.**

↓ Rows = Block time (flight length)

Longer flight = more cruise time = the lavatory consumes more water.

Going down → more coverage (greener)

→ Columns = Load Factor (how full)

A fuller aircraft carries more pax, so TAT boards more water — coverage drops.

Going right → less coverage (bluer)

Below 80%

**Blue** — water left over. Refinement opportunity.

80–99%

**Amber** — small margin. Confirm before adjusting.

100%+

**Green** — lavatory uses it all. TAT validated.

✈ Worked example — AUH → BAH (Bahrain), the highest-volume short flight

1Block time **1h08** — after 40 min of taxi, climb & descent, only **28 min of cruise** remain.

2In 28 min, at 4 min/use, 3 lavatories allow **18 uses** → the lavatory can consume **7.2 L**.

3TAT proposes boarding **13.2 L** for this flight.

4Coverage = 7.2 ÷ 13.2 = **54%** → **6.0 L left over** per flight × 1,020 flights/yr.

Coverage 54% → Blue cell. The 6.0 L leftover is the refinement Phase 1 confirms — part covers drinking water & crew, not just the lavatory.

The **bottom-left** (short + full flights) is the bluest — little time, lots of water. The **upper-right** (long flights) is all green. Short flights like BAH carry the most refinement potential.

💰 BLUE — ADDITIONAL MARGIN

20,732 flights

31.9% of NB network

$21,125 of additional refinement available. Flights where the lavatory consumes <80% of TAT water.

**Phase 1** confirms real onboard consumption.

🟠 AMBER — TRANSITION ZONE

23,820 flights

36.7% of NB network

$18,263 additional. Lavatory covers 80–99% of TAT water. Small margin — confirm with real data in Phase 1.

✅ GREEN — TAT VALIDATED

20,380 flights

31.4% of NB network

The lavatory consumes 100%+ of TAT water in cruise — the TAT figure is within what the aircraft physically uses.

💰 ADDITIONAL OPPORTUNITY — FULL NB NETWORK

$39,388

Refinement beyond TAT if we take the target to the lavatory's physical limit.

Blue: $21,125 · Amber: $18,263

🔴 SHORT-HAUL BAND — BLK ≤ 1H10

$4,890

5,848 flights where cruise time is too short for forecast consumption (BLK − 40 min overhead). Part covers consumption outside the lavatory (drinking water, crew). **Phase 1** confirms the real figure.

Break-even point — minimum BLK for the lavatory to validate TAT

LF 50% (88 pax eff.)

1.6h

Equil. above 1.6h BLK

LF 60% (105 pax eff.)

1.9h

Equil. above 1.9h BLK

LF 70% (123 pax eff.)

2.6h

Equil. above 2.6h BLK

LF 75% (132 pax eff.)

3.0h

Equil. above 3.0h BLK

LF 80% (140 pax eff.)

4.0h

Equil. above 4.0h BLK

LF 85% (149 pax eff.)

5.4h

Equil. above 5.4h BLK

LF 90% (158 pax eff.)

9.0h

Equil. above 9.0h BLK

LF 95% (167 pax eff.)

>9h

Never validates in NB range

LF 100% (176 pax eff.)

>9h

Never validates in NB range

Bars show % of BLK 0–6h · Point further right = later break-even = more blue zone

BLK × Load Factor map — Lavatory coverage (% of TAT water consumed in cruise)

| BLK | CRUISE | LF 50% | LF 60% | LF 70% | LF 75% | LF 80% | LF 85% | LF 90% | LF 95% | LF 100% |
|---|---|---|---|---|---|---|---|---|---|---|
| 0h30 | 0 min | 💰 0% | 💰 0% | 💰 0% | 💰 0% | 💰 0% | 💰 0% | 💰 0% | 💰 0% | 💰 0% |
| 0h45 | 5 min | 💰 12% | 💰 12% | 💰 12% | 💰 11% | 💰 11% | 💰 10% | 💰 10% | 💰 9% | 💰 9% |
| 1h | 20 min | 💰 60% | 💰 53% | 💰 46% | 💰 43% | 💰 40% | 💰 38% | 💰 36% | 💰 34% | 💰 33% |
| 1h15 | 35 min | 80% | 💰 68% | 💰 59% | 💰 55% | 💰 52% | 💰 49% | 💰 46% | 💰 44% | 💰 42% |
| 1h30 | 50 min | 100% | 85% | 💰 73% | 💰 69% | 💰 65% | 💰 61% | 💰 58% | 💰 55% | 💰 52% |
| 1h45 | 65 min | 114% | 97% | 84% | 💰 78% | 💰 74% | 💰 70% | 💰 66% | 💰 63% | 💰 60% |
| 2h | 80 min | 125% | 106% | 91% | 86% | 81% | 💰 76% | 💰 72% | 💰 68% | 💰 65% |
| 2h30 | 110 min | 135% | 114% | 99% | 93% | 87% | 82% | 💰 78% | 💰 74% | 💰 70% |
| 3h | 140 min | 146% | 123% | 107% | 100% | 94% | 89% | 84% | 80% | 💰 76% |
| 4h | 200 min | 156% | 132% | 114% | 107% | 101% | 95% | 90% | 86% | 82% |
| 5h | 260 min | 162% | 137% | 119% | 111% | 105% | 99% | 94% | 89% | 85% |
| 6h | 320 min | 167% | 141% | 122% | 114% | 108% | 102% | 96% | 91% | 87% |
| 8h | 440 min | 172% | 145% | 126% | 118% | 111% | 105% | 99% | 94% | 90% |

**Blue** Additional margin (<80%) **Amber** Transition (80–99%) **Green** TAT validated (≥100%)

Analysis by Block Time band

Short-haul

5,848 flights

$4,890 saving

100% blue 0% green

Medium

10,528 flights

$6,628 saving

65% blue 2% green

Long

7,540 flights

$5,349 saving

54% blue 14% green

Ultra (>3h)

41,016 flights

$22,521 saving

10% blue 47% green

**Executive summary:** With an average LF of **82.5%**, Etihad's NB network break-even sits around **3h** block time (at 4 min/use). **31.9% of flights** are in the blue zone — TAT boards more water than the lavatory can consume in the available cruise. Refining to the physical limit represents **$39,388** of additional opportunity on top of the WaterLine saving already calculated.

The **short-haul band (BLK ≤ 1h10)** is the densest blue — **5,848 flights** where cruise is too short for the forecast consumption. In the green zone (**31.4% of flights**), the TAT figure is already within what is consumable — physically validated by the lavatory.

**The WaterLine Validador does not invalidate the TAT model** — it maps where additional refinement opportunity exists in Phase 1. Parameters to confirm with Etihad Engineering: real time per use (currently **4 min**), volume per use (**400 ml**) and number of operational lavatories per flight (currently **3**).